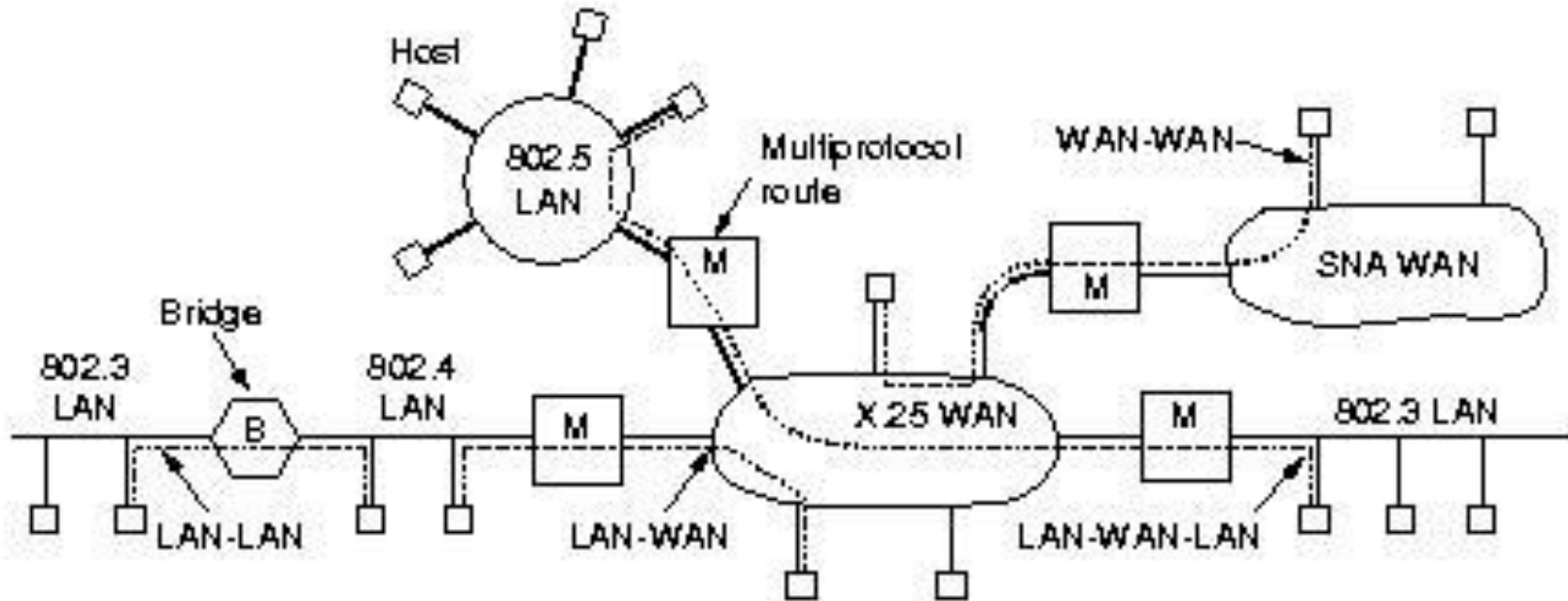


# Computer Networks

**Internetworking  
Internet Protocols**

# Internetworking

Connecting and working with multiple networks and protocols in order to deliver the packets.



**An example of mixing together multiple types of networks.**

# Internetworking

Internetworking deals with the issues of interconnecting multiple networks. Physical networks can be connected at several levels:

1. Repeaters operate at the physical layer (layer 1), copying signals from one LAN to another. They operate at the bit level, and have no notion of what the bits (or even frames!) mean.
2. Bridges operate at the data link layer (layer 2), copying frames from one LAN to another.
  - a. They perform store-and-forward packet switching, but use only level-2 (e.g. frame fields) information.
  - b. We've talked about these before in regard to the MAC layer, where we looked at spanning tree and source routing methods.
3. Routers operate at the network layer (level 3).
  - a. Similar to bridges in concept.
  - b. At the network layer, they are fully aware of different network technologies, and can problems as interconnect different between them.
4. Transport gateways connect two networks at the transport layer (level 4).
5. Application gateways operate at higher levels (level "7"). Application gateways can translate between OSI mail and SMTP (Internet) mail formats, for instance.

We've looked at some of these properties of differences:

| Item               | Some Possibilities                              |
|--------------------|-------------------------------------------------|
| Service Offered    | Connection-oriented versus connectionless       |
| Protocols          | IP, IPX, CLNP, Appletalk, DecNet, . . . .       |
| Addressing         | Flat (802) versus hierarchical (IP)             |
| Multicasting       | Present or absent (also broadcasting)           |
| Packet Size        | Every network has its own max                   |
| Quality of Service | May be present or absent - many different kinds |
| Error Handling     | Reliable, ordered, and unordered delivery       |
| Flow control       | Sliding window, rate control, other, none       |
| Congestion Control | Leaky bucket, choke packets, etc.               |
| Security           | Privacy rules, encryption, etc.                 |
| Parameters         | Different timeouts, flow specs, etc.            |
| Accounting         | By connect time, by packet, by byte, or none    |

Multiprotocol router use to connect two networks. If two networks have dissimilar network layers, the router may able to translate between the packet formats.

Multiprotocol Routers are packet switches that operate at the network layer (level 3). Operating at the network level gives routers increased flexibility compared to bridges in terms of:

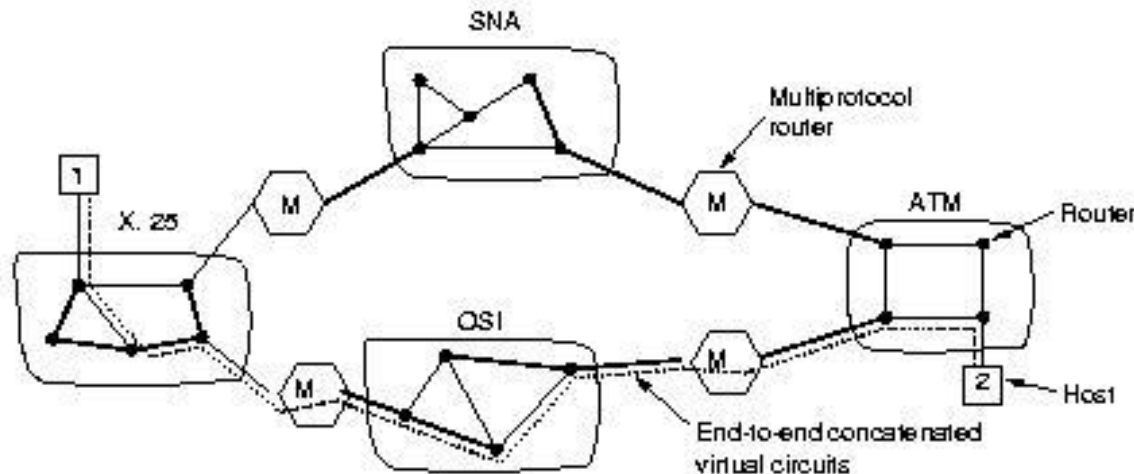
1. Translating addresses between dissimilar networks.
2. Fragmenting large packets for transmission across networks that carry only small maximum packet lengths.
3. Selecting an appropriate path through the subnet.
4. Enforcing policies (e.g., don't forward any local packets off of this network).

Two style of internetworking are possible: Connection-oriented concatenation virtual circuit subnets and datagram internet.

Internetworking in a connection-oriented , a connection to a host in a distant network is set up in a way similar to the way connection are normally established:

1. The sending host opens a virtual circuit as before, but now a circuit goes through router hops.
2. Any two neighboring routers at the internetworking level must be connected to a common network.
3. Regular router-based virtual circuits connect neighboring routers on the same physical network.
4. The end-to-end virtual circuit is a concatenation of individual virtual circuits through each of the networks along the path.

So each gateway/router maintains tables for each of the connections passing through it.



# Internetworking

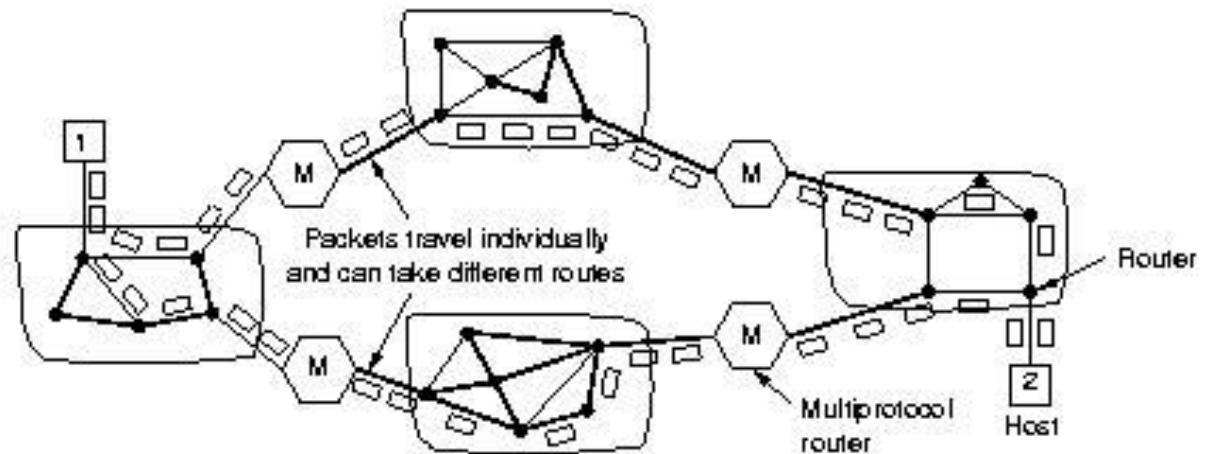
## Connectionless Internetworking

Connectionless internets operate just as connectionless networks.

- A host sends a packet to a neighboring router, which forwards it the next router, and so forth.
- Just as with connectionless networks, routers make only a best-effort attempt at delivering the packet.

### Datagrams

The Network layer puts datagrams on the subnet. See Figure



Issues that must be dealt with:

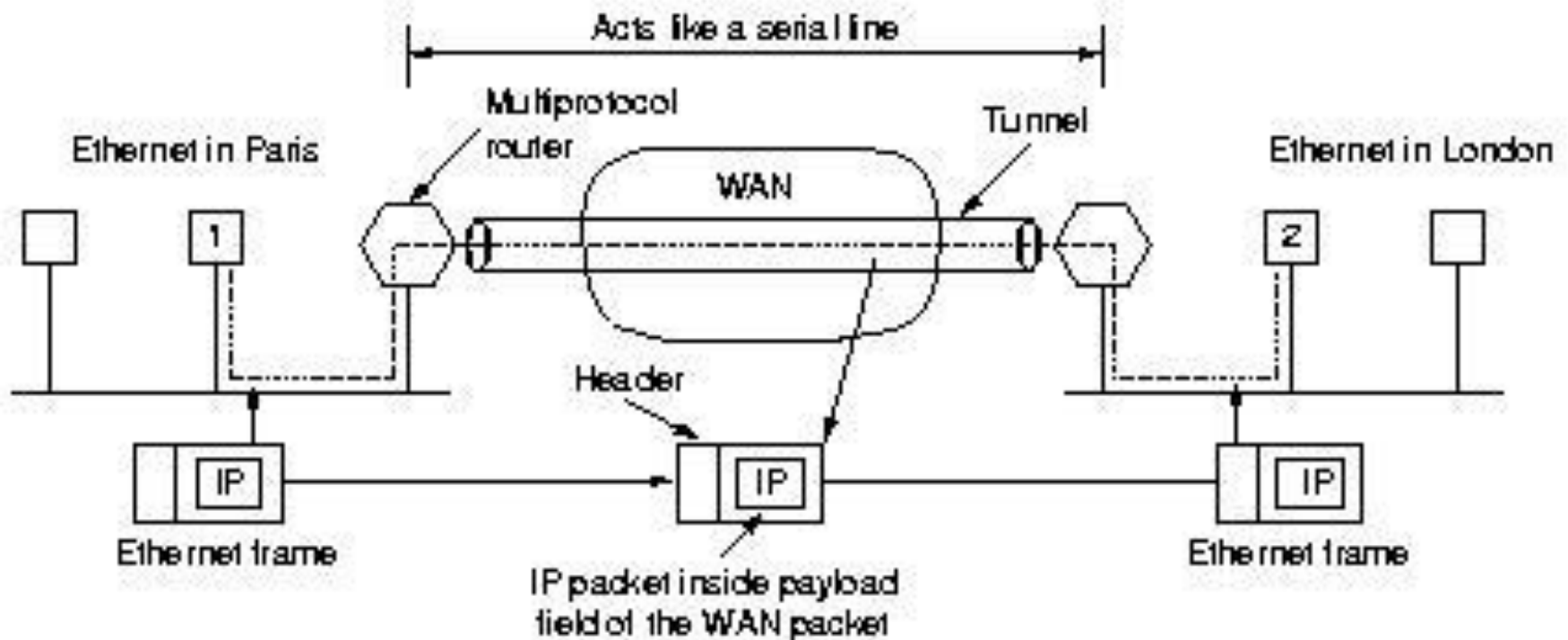
- Networks with different network protocols are tough to translate packets between them.
- Addressing - when adjacent networks have differing address schemes.

| Model                  | Advantages                                                                                                                                                                         | Disadvantages                                                                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Virtual Circuit</b> | <ul style="list-style-type: none"><li>• Buffers can be reserved in advance</li><li>• Sequencing guaranteed</li><li>• No delayed/duplicate packets</li></ul>                        | <ul style="list-style-type: none"><li>• Table space required</li><li>• Can't avoid congestion</li><li>• Impossible to implement if intervening network is unreliable</li></ul> |
| <b>Datagrams</b>       | <ul style="list-style-type: none"><li>• Can adapt to congestion</li><li>• Can handle router failures</li><li>• None of intervening networks need to be virtual circuits.</li></ul> | <ul style="list-style-type: none"><li>• Susceptible to congestion</li></ul>                                                                                                    |



Tunneling is a special case between two same-type networks across intervening foreign network(s).

- The whole packet is encapsulated in the protocol of the foreign network to be crossed, and then restored on the other side.
- This avoids, totally, trying to translate the packet.



Two-level of routing algorithms are used:

- Within each network an **interior gateway protocol** is used for routing.
- Between the networks, an **exterior gateway protocol** is used for routing.

- **Different packet sizes**

router may need to fragment incoming datagrams into smaller pieces, called segment or fragment, before transmitting on to the next network.

- **When to re-assemble**

- **At destination**

- Results in packets getting smaller as data traverses internet

- **Intermediate re-assembly**

- Need large buffers at routers
    - Buffers may fill with fragments
    - All fragments must go through same router
      - Inhibits dynamic routing

- **IP re-assembles at destination only**

- **IP fragmentation technique uses the following information in the IP header:**
  - **Data Unit Identifier (ID)**
    - Identifies end system originated datagram
      - Source and destination address
      - Protocol layer generating data (e.g. TCP)
      - Identification supplied by that layer
  - **Data length**
    - Length of user data in octets
  - **Offset**
    - Position of fragment of user data in original datagram
    - In multiples of 64 bits (8 octets)
  - **More flag**
    - Indicates that this is not the last fragment

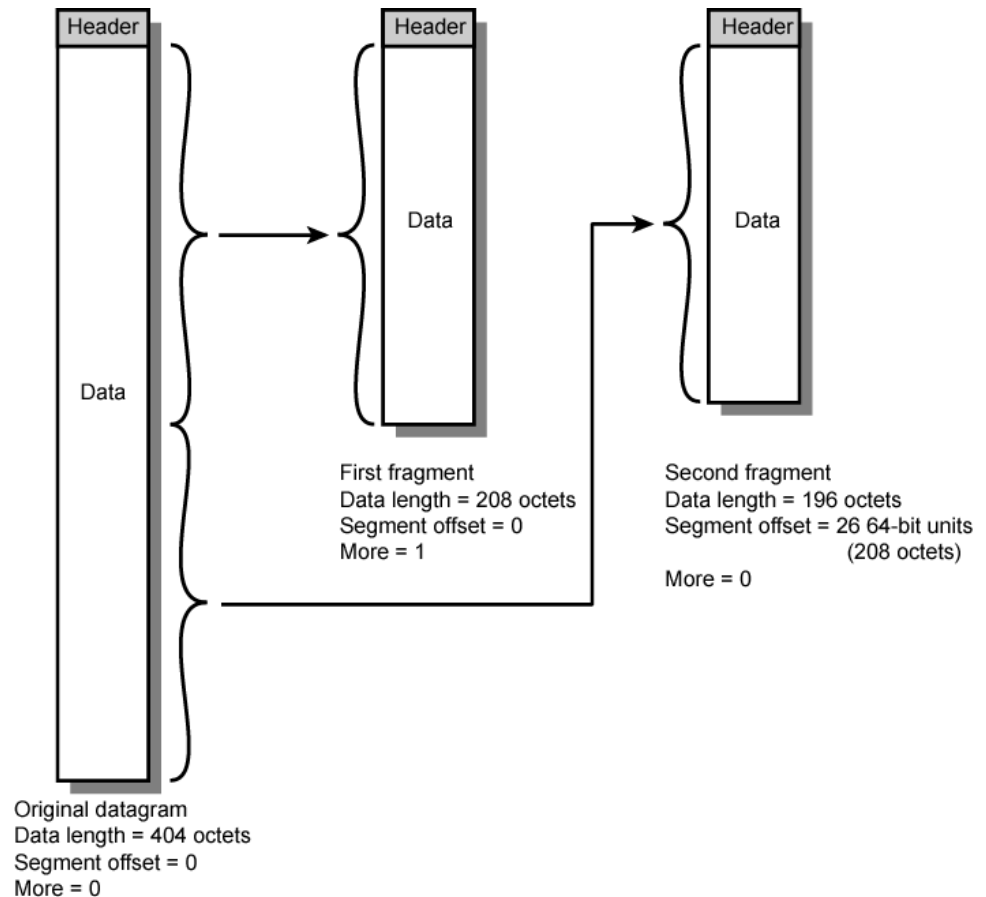
# Internetworking

## Fragmentation and Reassembly

The source end system creates a datagram with a Data Length equal to the entire length of the data filed, with Offset = 0, and a More Flag to 0.

**Fragment a long datagram into two pieces, an IP module in a router performs the following:**

1. Create two new datagrams and copy the header fields of the incoming datagram into both.
2. Divide the incoming user data filed into two approximately equal portions along a 64-bit boundary, placing one portion in each new datagram. The first portion must be a multiple of 64 bits (8 octets).
3. Set the Data Length of the first new datagram to the length of the inserted data, and set More Flag to 1. the Offset field is unchanged.
4. Set the data Length of the second new datagram to the length of the inserted data, and add the length of the first data portion divided by 8 to the Offset filed. The More Flag remains the same.



Generalized it to an n-way split.

To reassemble a datagram,

- There must be sufficient buffer space at the reassembly point.
- As Fragments with same ID arrive, their data fields are inserted in the proper position in the buffer until the entire data field is reassembled.
- Starting with Offset of zero and ending with data from a fragment with a false More flag.

**Example:** Suppose that host *A* is connected to a router *R*, and *R* is connected to host *B*. Suppose that a TCP message that contains 4440 bytes of data and 20 bytes of TCP header is passed to IP code at host *A* for delivery to *B*. Show the *Total length*, *DF*, *MF*, and *Fragment offset* fields of the IP header in each of the resulting fragments (packets) transmitted over the two links. Assume that link *A-R* can support a maximum frame size of 4504 bytes including 24-bytes frame header, link *R-B* can support a maximum frame size of 1024 bytes including 14-bytes frame header. Assume all IP headers are 20 bytes.

# Internetworking

## Fragmentation and Reassembly

**Solution.** The Subnet look like: A-----R-----B

TCP-PDU (protocol data unit) i.e. TCP message and TCP header is encapsulated into IP payload/data field

IP Data  $\leftarrow$  TCP-PDU = TCP Message + TCP Header = 4440 + 20 = 4460 bytes

IP Packet Size = Data + IP Header = 4460 + 20 = 4480 bytes.

**AR link:**

Capacity of AR link (Frame size) = 4504 bytes.

Payload/data of frame = 4504 – 24 = 4480 bytes.

So, Frame can encapsulate a IP packet into its data field.

**IP Header info:** Total length = 4480, Data length = 4460, DF = 0, MF = 0, Offset = 0.

**RB link:**

Capacity of RB link (Frame size) = 1024 bytes.

Payload/data of frame = 1024 – 14 = 1010 bytes.

So, Frame can not encapsulate a IP packet into its data field. Router fragments the IP packet.

Maximum IP packet/fragment size for this link = 1010 bytes. Or

Maximum IP data length = IP Size - IP header = 1010 – 20 = 990 bytes.

IP data length = 984 bytes. ( divisible by 8).

There are four fragments for better utilization of RB link.

Total Length = 984 + 20

Data Length = 984

DF = 0

MF = 1

Offset = 0

Total Length = 984 + 20

Data Length = 984

DF = 0

MF = 1

Offset = 123

Total Length = 984 + 20

Data Length = 984

DF = 0

MF = 1

Offset = 246

Total Length = 984 + 20

Data Length = 984

DF = 0

MF = 1

Offset = 369

TL = 524 + 20

DL = 524

DF = 0

MF = 0

offset = 492

# INTERNET PROTOCOL (IPv4)

## INTERNET PROTOCOL (IP)

The goal of IP is to interconnect networks of diverse technologies and create a single, virtual network to which all hosts connect.

- The Internet Protocol (IP) is a network layer protocol.
- IP provides connectionless, best-effort delivery service to the layers above it.
- Hosts communicate with other hosts by handing datagrams to the IP layer;
- The sender doesn't worry about the details of how the networks are actually interconnected.
- IP defines a universal packet called an Internet Datagram.

All Hosts and gateways process IP datagrams.



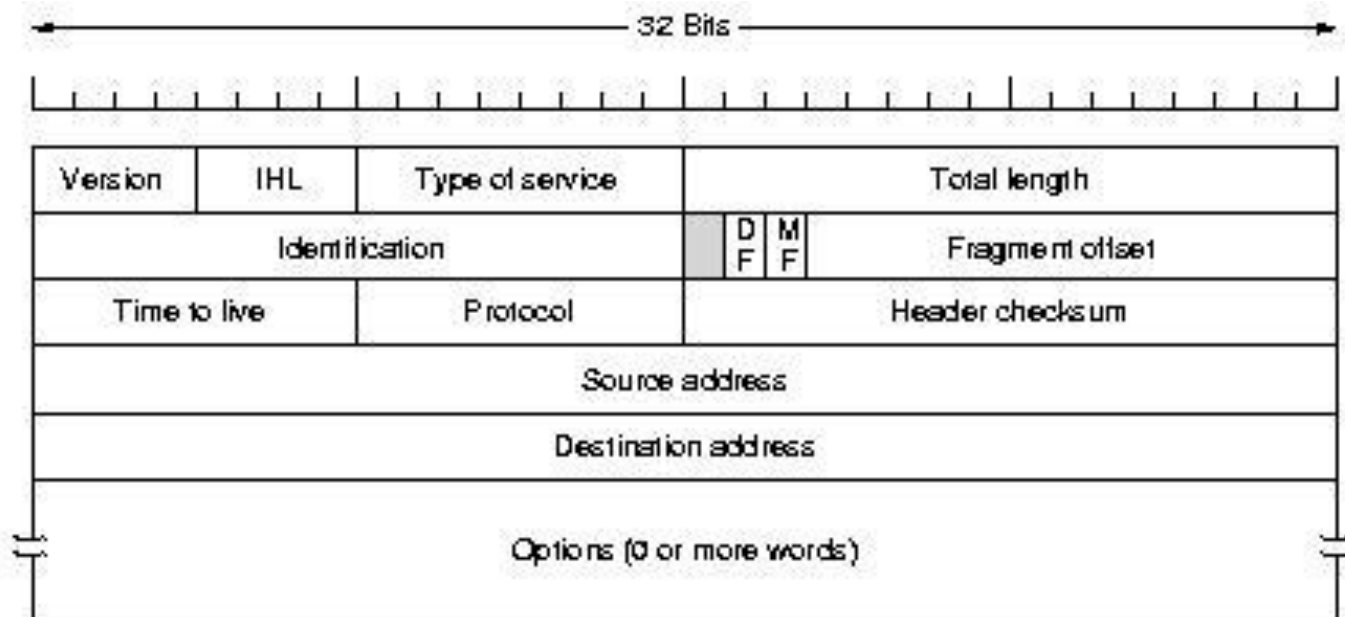
# The IPv4 Protocol

## 1. Version number (4-bits):

- The current protocol version is 4.
- Including a version number allows a future version of IP be used along side the current version, facilitating migration to new protocols.

## 2. Header length (4-bits):

- Length of the datagram header (excluding data) in 32-bit words.
- The minimum length is 20 bytes (IHL=5), but can be up to 60 bytes (IHL=15) if options are used.
- In practice, the length field is used to locate the start of the data portion of the datagram.



# The IPV4 Protocol

## 3. Type-of-service (8-bits):

A hint to the routing algorithms as to what type of service we desire.

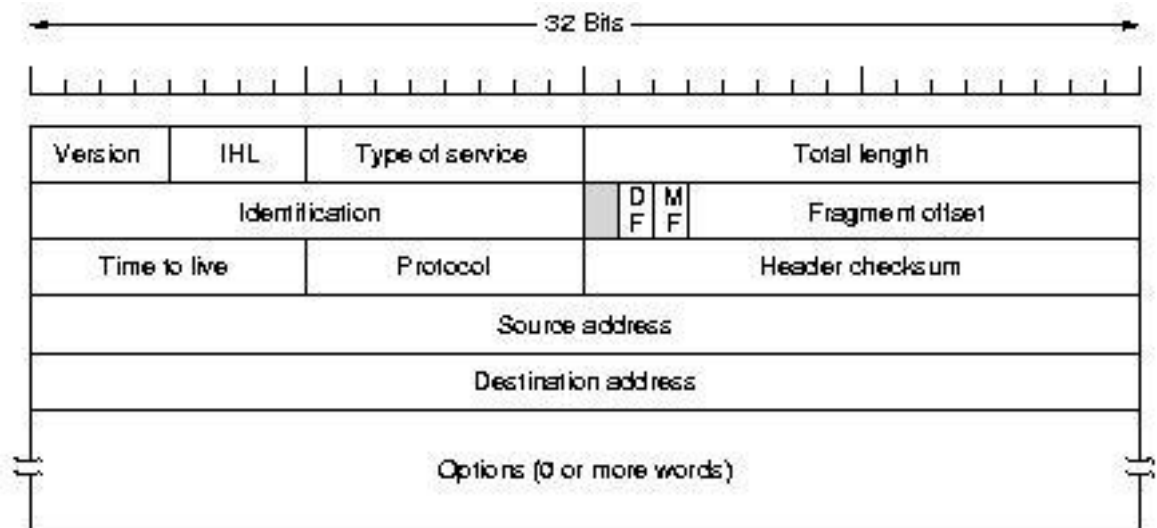
**Precedence** (3-bits): A priority indication, where 0 is the lowest and means normal service, while 7 is highest and is intended for network control messages (e.g., routing, congestion control).

**Delay** (1-bit): An Application can request low delay service (e.g., for interactive use).

**Throughput** (1-bit): Application requests high throughput.

**Reliability** (1-bit): Application requests high reliability.

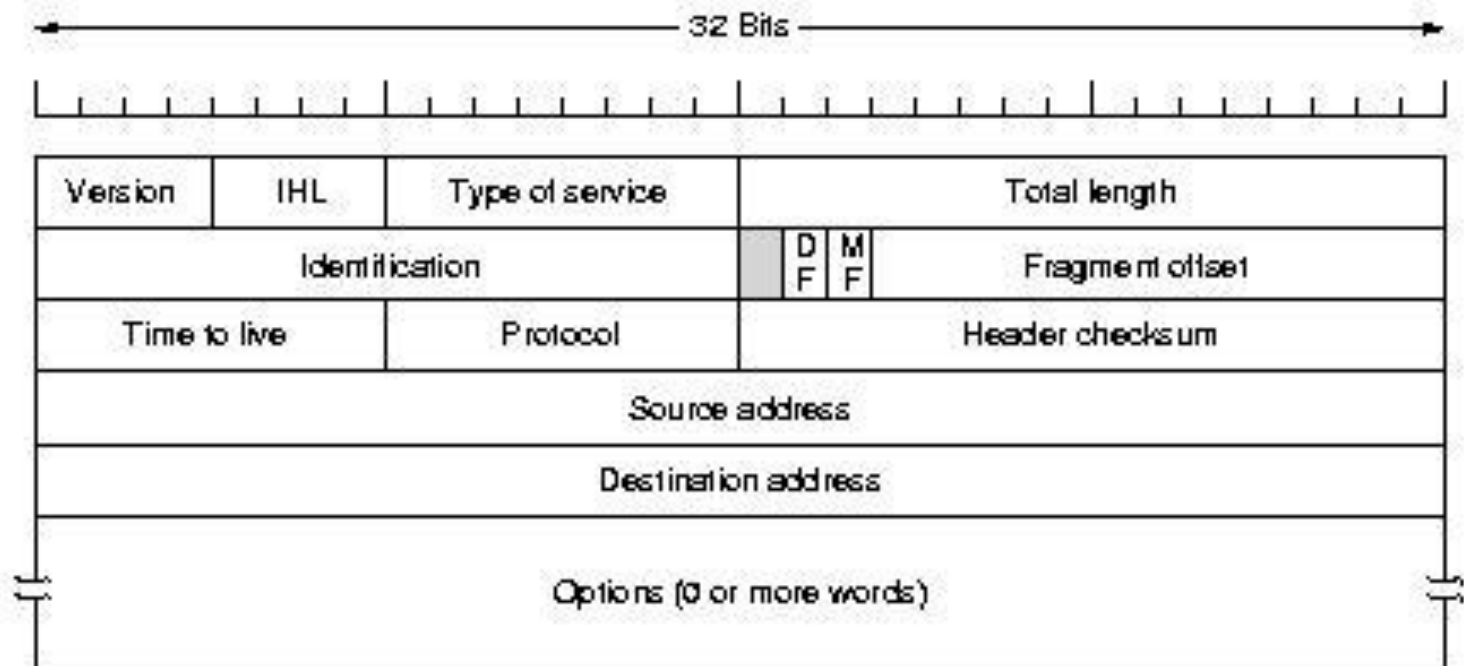
Note: These last three TOS bits will generally be mutually exclusive. Does setting the low-delay bit guarantee getting such service? No. The type-of-service field is meant as a request or hint to the routing algorithms, but does not guarantee that your request can be honored (e.g., there may not be a low-delay path available).



# The IPv4 Protocol

## 4. Total length (16-bits):

Total length of the IP datagram (in bytes), including data and header. The size of the data portion of the datagram is the total length minus the size of the header.



# The IPv4 Protocol

## 5 - 8. Identification (16-bits), Flags (unused, DF,MF)(3-bits), Fragment offset (13-bits):

These three fields are used for fragmentation and reassembly.

- Gateways along a path are free to fragment datagrams as needed; hosts are required to reassemble fragments before passing complete datagrams to the higher layer protocols.
- Each fragment contains a complete copy of the original datagram header plus some portion of the data.
- A receiving host must match arriving fragments with the proper original datagram.
- These fragments may be out of order and interleaved with other fragments.
- All fragments of a datagram will have the same source and destination IP address.
- But, other datagrams between those two machines will share these fields as well, so this is not enough.

# The IPv4 Protocol

## 5 - 8. Identification (16-bits), Flags (3-bits), Fragment offset (13-bits) (Continued):

The identification field uniquely identifies fragments of the same original datagram.

Whenever a host sends a datagram, it sets the identification field of the outgoing datagram and increments its local identification counter.

The offset field shows order of the fragments.

When a gateway fragments a datagram, it sets the offset field of each fragment to reflect at what data offset with respect to the original datagram the current fragment belongs.

Fragmentation occurs in 8-byte chunks, so the offset holds the “chunk number”.

Gateways can further fragment fragments!

A 400-byte fragment having an offset of 300 chunks could be split into two 200-byte fragments having offsets of 300 and 325 chunks, respectively.

# The IPv4 Protocol

We need to know when we've received all of the fragments. To help with this, the flags field may contain:

A **Don't Fragment** indication (set by host, honored by gateways). (A 1-bit flag.)

The **More Fragments** field indicates that another fragment follows this one. This fragment is not the last fragment of the original datagram.

An unfragmented datagram has an offset of 0, and a More Fragment bit of 0.

The last fragment of a fragmented datagram contains More Fragment = Clear and the Offset non-zero.

## Note:

The total length field of the IP header refers to the current datagram, not the original.

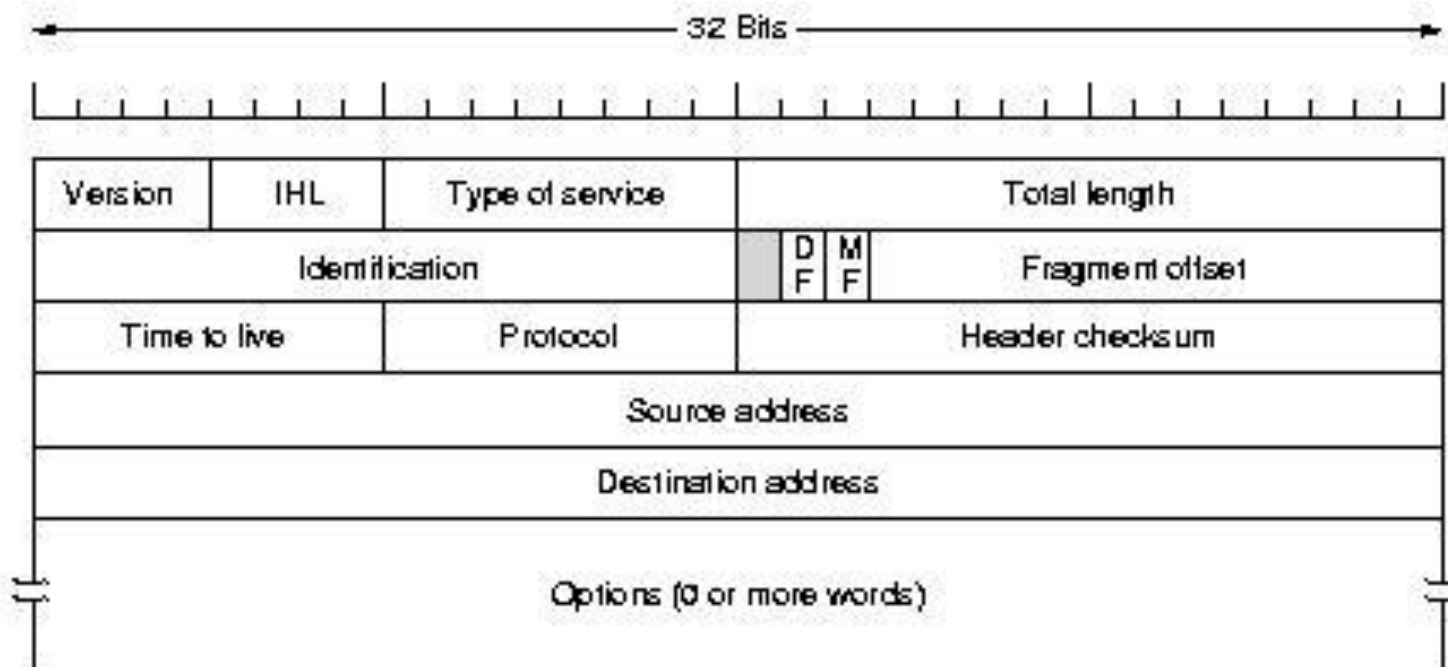
Thus, the More Fragment bit is needed in order for the recipient host to determine when it has all fragments of a datagram.

# The IPv4 Protocol

## 5 - 8. Identification (16-bits), Flags (3-bits), Fragment offset (13-bits) (Continued):

### Example:

Original Frame: IHL = 5, Length = 656, Fragment Offset = 0, More = 0  
Fragment 1: IHL = 5, Length = 252, Fragment Offset = 0, More = 1  
Fragment 2: IHL = 5, Length = 252, Fragment Offset = 29, More = 1  
Fragment 3: IHL = 5, Length = 192, Fragment Offset = 58, More = 0



# The IPv4 Protocol

## 9. Time-to-live (8-bits):

- A counter that is decremented by each gateway.
- Should this hop count reach 0, discard the datagram.
- Originally, the time-to-live field was intended to reflect real time.
- In practice, it is now a hop count.
- The time-to-live field squashes looping packets.
- It also guarantees that packets don't stay in the network for longer than 255 seconds, a property needed by higher layer protocols that reuse sequence numbers.

## 10. Protocol (8-bits):

- What type of data the IP datagram carries (e.g., TCP, UDP, etc.).
- Needed by the receiving IP to know the higher level service that will next handle the data.



# The IPv4 Protocol

## 11. Header Checksum (16-bits):

A checksum of the IP header (excluding data).

The IP checksum is computed as follows:

- Treat the data as a stream of 16-bit words (appending a 0 byte if needed).
- Compute the 1's complement sum of the 16-bit words. Take the 1's complement of the computed sum.

This checksum is much weaker than the CRCs we have studied.

But, it has the property that the order in which the 16-bit words are summed is irrelevant.

We can place the checksum in a fixed location in the header, set it to zero, compute the checksum, and store its value in the checksum field.

On receipt of a datagram, the computed checksum calculated over the received packet should be zero.

Check summing only the header reduces the processing time at each gateway, but forces transport layer protocols to perform error detection (if desired).

The header must be recalculated at every router since the `time_to_live` field is decremented.

# The IPV4 Protocol

## 12. Source address (32-bits):

Original sender's address. This is an IP address, not a MAC address.

## 13. Destination address (32-bits):

Datagram's ultimate destination.

Note: When a gateway forwards a frame to another gateway, it forwards an Ethernet frame.

The IP embedded datagram contains the source of the original sender (not the forwarding gateway) and the destination address of the ultimate destination.

# The IPv4 Protocol

## 14. IP Options

IP datagrams allow the inclusion of optional, varying length fields that need not appear in every datagram. We may sometimes want to send special information, but we don't want to dedicate a field in the packet header for this purpose.

Options start with a 1-byte option code, followed by zero or more bytes of option data.

The option code byte contains three parts:

**copy flag (1 bit):** If 1, replicate option in each fragment of a fragmented datagram. That is, this option should appear in every fragment as well. If 0, option need only appear in first fragment.

**option class (2 bits):** Purpose of option:

**0 = network control**

**1 = reserved**

**2 = debugging and measurement**

**3 = reserved**

| Option                | Description                                        |
|-----------------------|----------------------------------------------------|
| Security              | Specifies how secret the datagram is               |
| Strict source routing | Gives the complete path to be followed             |
| Loose source routing  | Gives a list of routers not to be missed           |
| Record route          | Makes each router append its IP address            |
| Timestamp             | Makes each router append its address and timestamp |

**option number (5 bits):** A code indicating the option's type. See Figure for these.

# IP ADDRESSES

Addresses consist of compact, 32-bit identifiers. Internet software translates names into addresses and addresses into names; lower protocol layers always uses addresses rather than names.

Internet addresses are hierarchical, consisting of two parts:

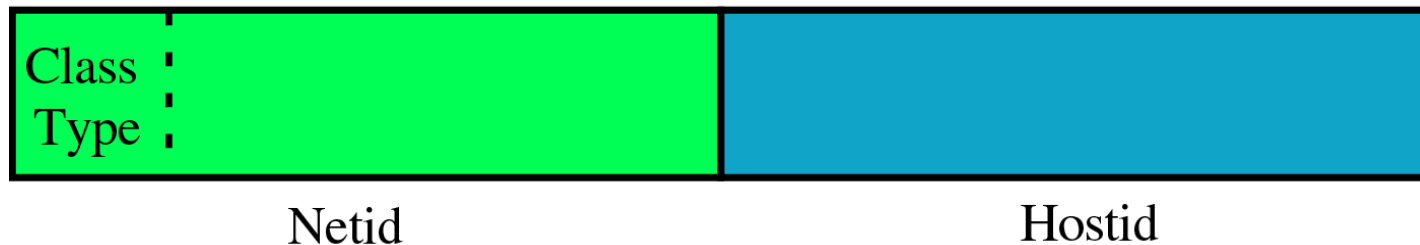
- **Network:** The network part of an address identifies which network a host is on. Conceptually, each LAN has its own unique IP network number.
- **Host:** The host part of an address identifies which host on that network.

Every host and router on the Internet has an IP address.

No two machines on the Internet have the same IP address.

Internet address are four bytes in size.

An Internet address is made of four bytes  
(32 bits) that define a host's connection  
to a network.



# IP Address Classes

The Internet designers were unsure whether the world would evolve into a few networks with many hosts (e.g., large networks), or many networks each supporting only a few hosts (e.g., small networks).

Thus, Internet addresses handle both large and small networks.

IP addresses were divided onto the five classes.

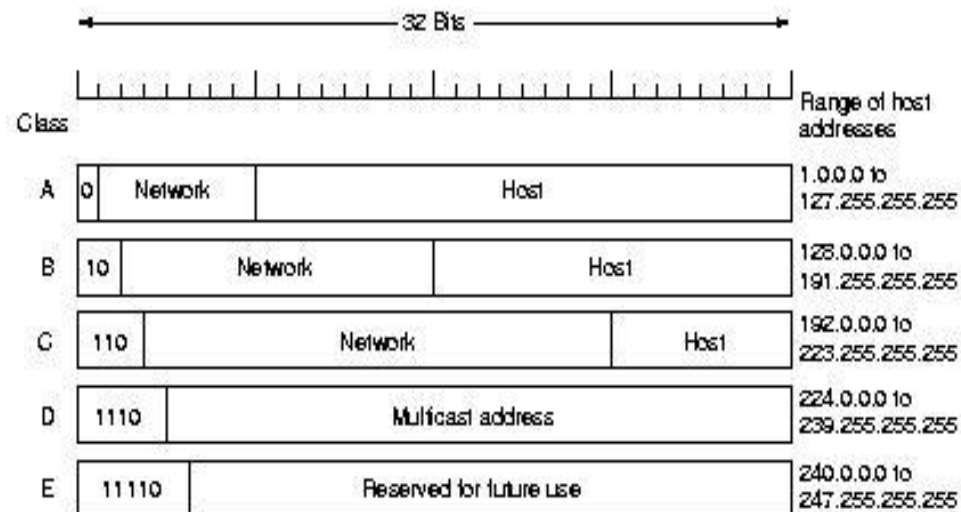
**Class A** addresses start with a '0' in the most significant bit, followed by a 7-bit network address and a 24-bit host part.

**Class B** addresses start with a '10' in the two most significant bits, followed by a 14-bit network number and a 16-bit host part.

**Class C** addresses start with a '110' in the three most significant bits, followed by a 21-bit network number and an 8-bit host part.

**Class D** addresses start with a '1110' in the four most significant bits, followed by a 28-bit group number. Used for multicast.

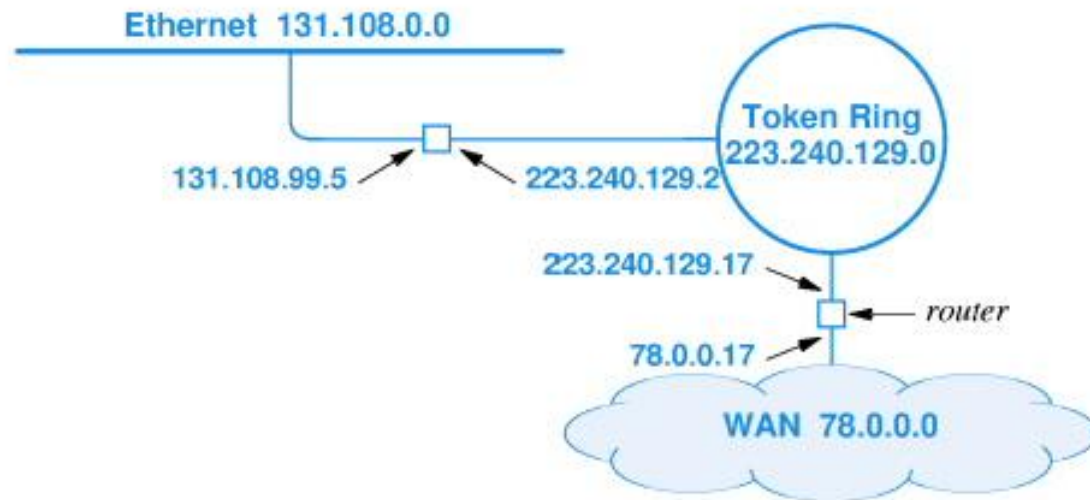
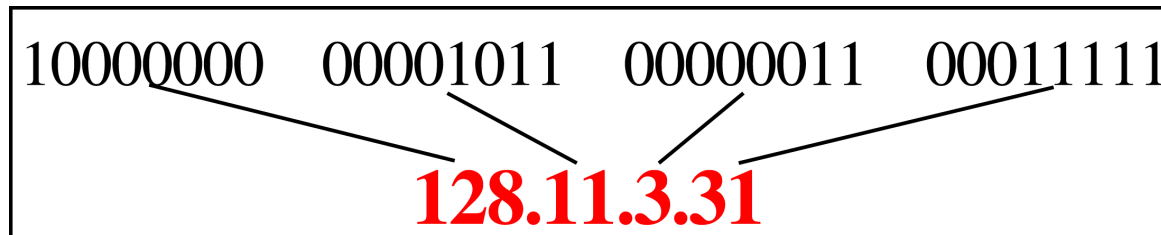
**Class E** addresses start with a '11110' and are reserved for future use.



# IP Addresses

IP addresses usually written in ***dotted decimal notation***.

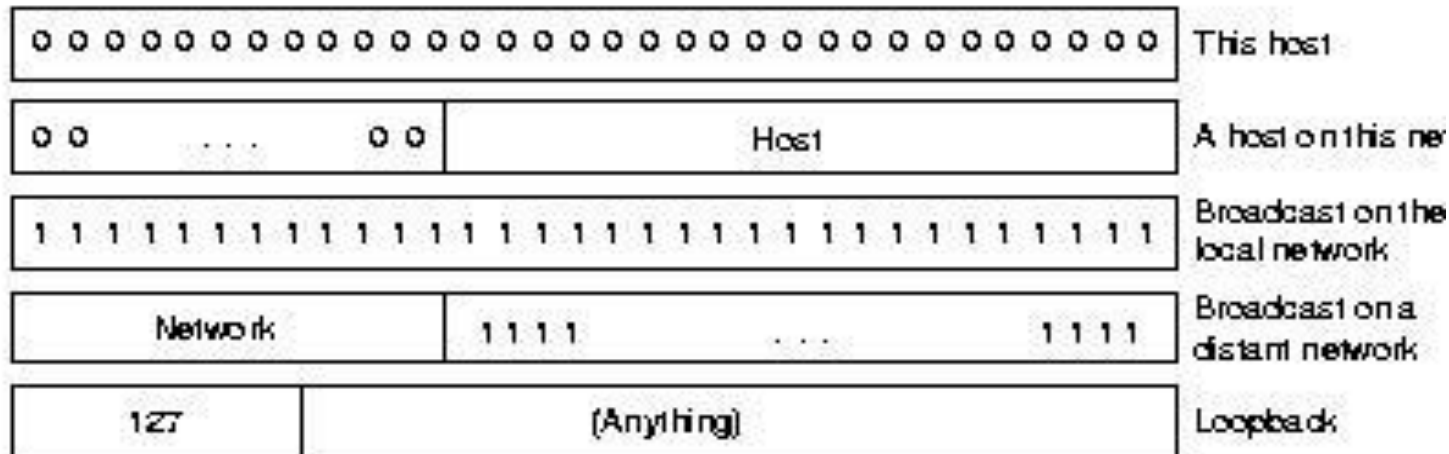
Each of four bytes is written in decimal from 0 to 255.



# Special IP Addresses

The use of fixed-sized IP addresses makes the routing operation efficient.

Registration of addresses is through the NIC (Network Information Center.)



- The IP address 0.0.0.0 is used by a hosts when they are booted.
- IP addresses with 0 as network number refer to the current network.
- The address consisting of all 1s allow broadcasting on the local network.
- The addresses with a proper network number and all 1s in the host field allow machine to send broadcast packets to distant LANs anywhere in the Internet.
- All the addresses of form 127.xx.yy.zz are reserved for loop back testing. Allows packets to be sent to the local network without the sender knowing its number.

# IP Address

Note: Internet addresses refer to network connections rather than hosts.

- a) Gateways, for instance, have two or more network connections and each interface has its own IP address.
- b) There is not a one-to-one mapping between host names and IP addresses.

Internet addresses are hierarchical.

- a) Datagrams are initially routed only by network number.
- b) Only the gateway connected to the destination network uses the local (host) part while performing the routing operation.

What happens to a host's internet address if that host moves from one network to another?

- a) Its Internet address must change.
- b) It's important to distinguish between a machine's name and its address.
- c) Physical (Ethernet) address is constant, network (IP) address may change.



# Subnets

This usage of “Subnets” is different from that we used before to define the routers and lines in a network.

## Goals:

- We want to be able to reduce the number of networks seen by the outside world;
  - We want to simplify the management of those many networks within the organization;
1. A large organization or campus might have 30 or more LANs (one for each department).
  2. An organization will probably have only a single connection to the rest of the Internet.
  3. In order for every local host to be able to communicate with other Internet machines, routing entries for each of the 30 networks must exist in the core gateways.
  4. In order for other sites to be able to respond to our queries, they must be able to route packets back to us.
  5. Wouldn't it be nice if we only needed to advertise a single network number for all 30 networks?

## The Answer:

- Subnet addressing is a technique that allows a set of multiple, interconnected networks to be covered by a single IP network number.
- IP addresses have a well-defined structure that allows a gateway to extract the network portion of an address by simply looking at its class and an optional netmask.

# Subnets

With subnetting, the local part of an IP address is further subdivided into a network and a host part:

Consider two addresses 128.204.2.29 and 128.204.3.109.

Are they on the same network?

**NO.**

- They refer to hosts on the same network address (128.204), but they can actually be on **different** ethernet networks connected by a bridge.
- To do this, we divide the local part (the two bytes to the right of 128.204) into a 1-byte network part and a 1-byte host part.
- When sending data to 128.204.3.109 local gateways first route datagrams to the (sub)network 128.204.3 rather than (IP network) 128.204.
- 128.204.2 and 128.204.3 are distinct (sub)networks.
- To the outside world, there is only a single network 128.204.
- Each of the individual networks is called a subnet.

# Subnets

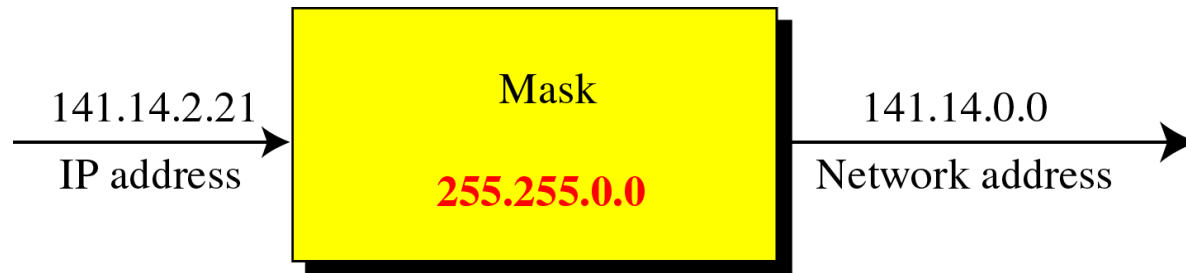
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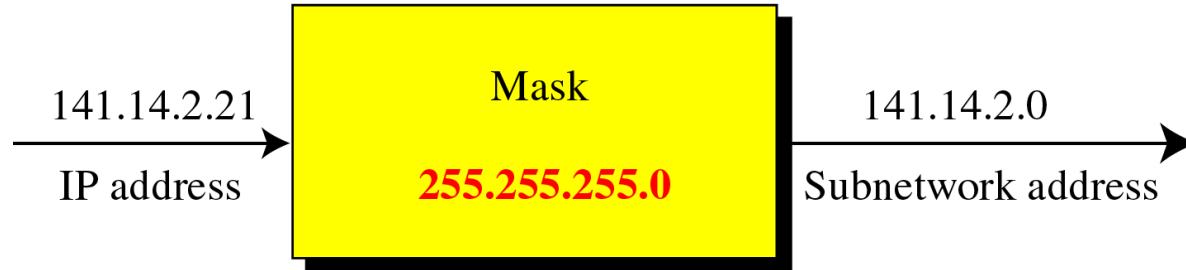
**YES.**

- They refer to hosts on the same network address (128.204), but they can actually be on **the same** ethernet.
- To do this, we divide the local part (the two bytes to the right of 128.204) into a 7-bit network part and a 9-bit host part.
- Our example above is a Class B address; the technique applies also to Classes A and C.

# Subnet Masking

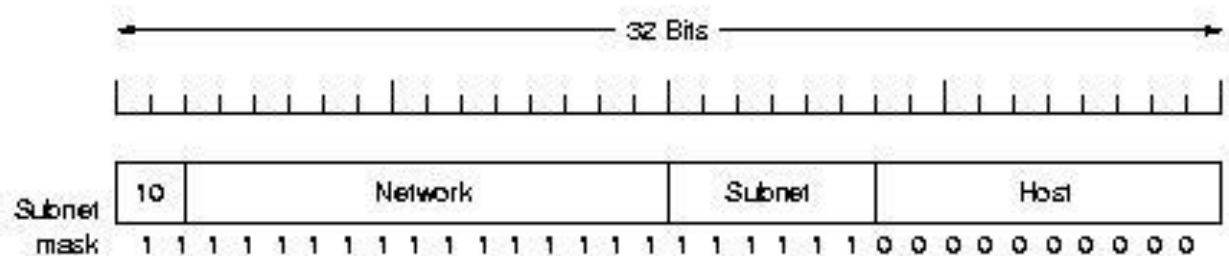


a. Without subnetting



b. With subnetting

# Subnetting



To implement subnetting, hosts and gateways use a subnet mask to extract the network part (network + subnet) of an IP address.

This mask can be seen in above Figure. In this example, 6 bits are reserved for subnet, and 10 bits for host.

Subnet masks are also written in dotted decimal notation, with the addition of a slash followed by a number of bits in the network + subnet part.

Subnet mask of above example can be written as 255.255.252.0 or alternative notation is /22 to indicate that the subnet mask is 22 bits long.

e.g. A packet addressed to 130.50.16.6 and arriving at the main router is ANDed with the subnet mask 255.255.252.0/22 to give the address 130.50.12.0.

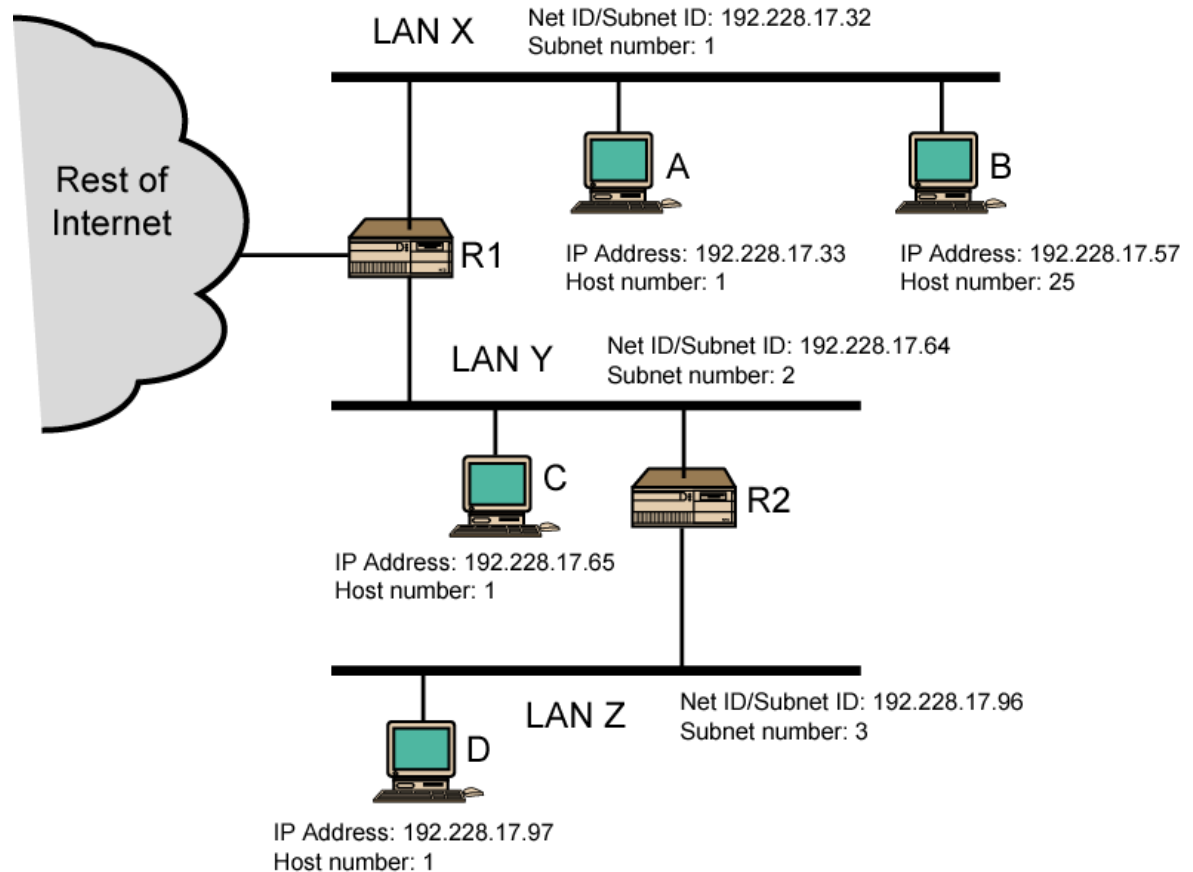
This address is looked up in the routing table to find out which output line to use to get the router for subnet 3.

# Routing using Subnets

Both the R1 and R2 configured with subnet mask 255.255.255.224.

If a datagram with a destination address 192.228.17.57 arrives at R1 from either rest of the Internet or from LAN Y, R1 apply the subnet mask that is subnet 1, which is LAN X, and so forwarded datagram to LAN X.

If the datagram with the same destination address arrives at R2 from LAN Z, R2 applies the mask and then determines from its forwarding database (table) that datagram destined for subnet 1 should be forwarded to R1.



|                                                                      |                                     |                 |
|----------------------------------------------------------------------|-------------------------------------|-----------------|
| IP Address                                                           | 11000000.11100100.00010001.00111001 | 192.228.17.57   |
| Subnet mask                                                          | 11111111.11111111.11111111.11100000 | 255.255.255.224 |
| Bitwise AND of address and mask<br>(resultant network/subnet number) | 11000000.11100100.00010001.00100000 | 192.228.17.32   |
| Subnet number                                                        | 11000000.11100100.00010001.001      | 1               |
| Host Number                                                          | 00000000.00000000.00000000.00011001 | 25              |

# CIDR – Classless InterDomain Routing

- There a single routing table for all networks consisting of an array of IP address, subnet mask, outgoing line.
- When a packet comes, its destination IP address is first extracted. Then
- The routing table is scanned entry by entry, masking the destination address.
- Compare it to the table entry looking for a match.
- If matched, then use the outgoing line of that entry.
- If multiple entry (with different subnet mask lengths) match, in this case the longest mask is used.
- E.g. if there is a match for a /20 mask and a /24 mask, the /24 entry is used.

# CIRD – Classless InterDomain Routing:

## IP Address Assignment:

To make the forwarding algorithms easier to understand.

Let us consider an example in which millions of addresses are available starting at 194.24.0.0

- Suppose that there are three universities – JNU, DU and JMI request 2048, 4096 and 1024 addresses respectively and in that order.
- JNU assigned the addresses 194.24.0.0 to 194.24.7.255, along with mask 255.255.248.0. (in 2048 byte boundary e.g.  $2^{11}$ )
- DU asks for 4096 addresses. Since a block of 4096 addresses must lie on a 4096 bytes boundary e. g.  $2^{12}$ , they can not be given addresses starting at 194.24.8.0. instead, they get 194.24.16.0 to 194.24.31.255 along with subnet mask 255.255.240.0.
- JMI asks for 1024 addresses (in 1024 byte boundary e.g.  $2^{10}$ ), assigned the addresses 194.24.8.0 to 194.24.11.255, along with subnet mask 255.255.252.0.

| University | First address | Last address  | How many | Written as     |
|------------|---------------|---------------|----------|----------------|
| JNU        | 194.24.0.0    | 194.24.7.255  | 2048     | 194.24.0.0/21  |
| JMI        | 194.24.8.0    | 194.24.11.255 | 1024     | 194.24.8.0/22  |
| Available  | 194.24.12.0   | 194.24.15.255 | 1024     | 194.24.12.0/22 |
| DU         | 194.24.16.0   | 194.24.31.255 | 4096     | 194.24.16.0/20 |



# CIRD – Classless InterDomain Routing:

Question: Suppose a router has built up the routing table shown in table. The router can deliver packets directly over *interfaces* 0 and 1, or it can forward packets to routers *R2*, *R3*, or *R4*. Assume the router does the longest prefix match (longest subnet mask) in case of more than one subnet addresses matched after performed masking on a given IP address using all subnet masks. If no subnet addresses matched, use default entry. Describe what the router does (i.e. what is next hop) with a packet addressed to each of the following destinations:

- 128.96.171.92
- 128.96.167.151
- 128.96.163.151
- 128.96.169.192

- Extract the *host id* from above IP addresses.

128.96.171.92 & 255.255.254.0 = 128.96.170.0

128.96.171.92 & 255.255.252.0 = 128.96.168.0

Longest mask 254 – use interface 0.

Class B address – net id 16 and local 16

254 = 11111110 - subnet id – 7 bit and host – 9 bit

171.92 = 10101011 . 01011100 - host id = 101011100=348

Do the same for all other three addresses.....

| Subnet Address | Subnet Mask   | Next Hop    |
|----------------|---------------|-------------|
| 128.96.170.0   | 255.255.254.0 | Interface 0 |
| 128.96.168.0   | 255.255.254.0 | Interface 1 |
| 128.96.166.0   | 255.255.254.0 | R2          |
| 128.96.164.0   | 255.255.252.0 | R3          |
| default        |               | R4          |

Question: An organization has a class C network 200.1.1 and wants to form subnets for four departments *D1*, *D2*, *D3* and *D4*, with hosts as follows:

- D1* 72 hosts
- D2* 35 hosts
- D3* 20 hosts
- D4* 18 hosts

There are 145 hosts in all.

Give a possible arrangement of subnet masks to make this possible.

Suggest what the organization might do if department *D4* grows to 34 hosts and give a possible arrangement of subnet masks.

# Internet Control Protocols

## INTERNET CONTROL MESSAGE PROTOCOL (ICMP)

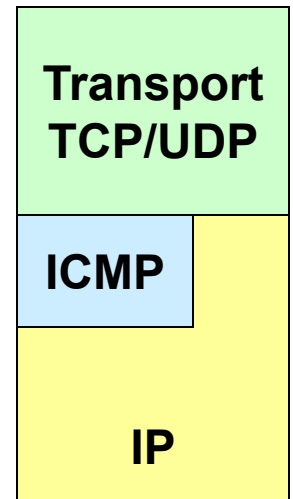
The Internet Control Message Protocol (ICMP) allows gateways and hosts to send network control information to each other.

From a layering point of view, ICMP is a separate protocol that sits above IP and uses IP to transport messages.

In practice, ICMP is an integral part of IP and all IP modules must support the ICMP protocol.

ICMP datagrams are encapsulated within IP datagrams and processed by IP in the same way as TCP and UDP datagrams;

if special processing is needed, the IP type-of-service (TOS) field could be used.



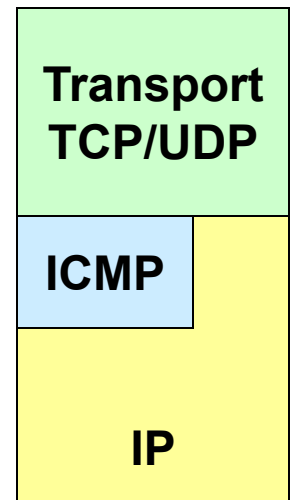
# INTERNET CONTROL MESSAGE PROTOCOL (ICMP)

There are two general types of ICMP messages:

**Information messages**, where a sender sends a query to another machine (either host or gateway) and expects an answer. For example, a host might want to know if a gateway is alive.

**Error indication messages**, where the IP software on a host or gateway has encountered a problem processing an IP datagram. For example, it may be unable to route a datagram to its destination, or it may have had to drop a frame.

There are a number of message types of which we will talk about only a few:



# INTERNET CONTROL MESSAGE PROTOCOL (ICMP)

## Echo Requests (Reply)

The ICMP echo request and echo reply messages are useful for network debugging.

If machine A sends an echo request message to machine B, machine B is required to respond with an ICMP echo reply.

Most systems supply an application program that sends and receives ICMP echo messages.

In UNIX, the program ping allows a user to check whether a machine is reachable and functioning.

Because ICMP messages are handled just like other IP datagrams, ICMP echo messages test the reachability of any host. Also, because ICMP is an integral part of IP, all hosts and gateways must implement ICMP.

## Timestamp Messages (Request and Reply)

ICMP timestamp messages are used to estimate the transmission delays between machines and to synchronize clocks:

Including both the receive and transmit timestamp allows the sending host to determine the fraction of time spent transmitting vs. processing the request.

# INTERNET CONTROL MESSAGE PROTOCOL (ICMP)

When an IP module encounters an error while processing a datagram, it sends an ICMP error message back to the original sender of the datagram. Errors include:

**Destination Unreachable:** When a gateway cannot route a datagram (e.g., it doesn't have an appropriate route in its local table), it discards the message and returns an ICMP "destination unreachable" message to the sending host. In effect, the host needs different routing or needs to try again later.

**Time Exceeded:** As a datagram is processed, gateways decrement its time-to-live (TTL) field. If the TTL value reaches 0, the gateway discards the datagram and sends a time exceeded message to the sender. The data portion of the message includes part of the offending datagram's header.

**Parameter Problem:** When a host or gateway encounters a problem parsing an IP datagram, it returns a parameter problem message to the datagram's sender:

**Source Quench:** When a gateway becomes congested and runs out of buffer space, it may discard a datagram and return a source quench message (choke packet). Source quench messages are used to request that the sender reduce the rate at which it is sending datagrams.

# MAPPING BETWEEN INTERNET AND PHYSICAL ADDRESSES

Suppose we have two machines A and B connected to the same network, and A wants to send an internet datagram to B.

A must know B 's data link layer (MAC) address in order to send frames to B.

The problem of mapping Internet addresses to physical addresses is known as the address resolution problem.

1. Each e-net device has its own unique number. Change the card and you change its physical address.
2. Physical address are 6 bytes long, too large to multiplex within an Internet address.
3. New machines can be added to the network with no disruption of service.
4. But, adding new hosts should not require reconfiguring existing hosts to inform them of the new machine.

# Address Resolution Protocol (ARP)

The Address Resolution Protocol (ARP) is a protocol that allows hosts to dynamically map Internet addresses to physical addresses:

1. The requesting machine only needs to know the target machine's IP address.
2. It sends out a special ARP request frame using the Ethernet's broadcast capability. Thus, every machine on the LAN will receive the ARP request.
3. The ARP request asks `what is the Ethernet address of Internet address A.B.C.D'?
4. Each machine receives a copy of the broadcast message, and the machine having the desired IP address responds with its Ethernet address.

Of course, a machine doesn't send out an ARP packet each time it wishes to send an IP datagram.

Instead, each machine maintains a cache of recently used mappings, and an ARP request is only sent if the desired mapping is not already in the cache.

# ARP

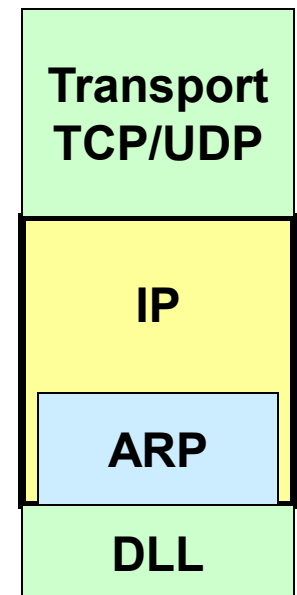
ARP request packets also contain the sender's IP and Ethernet address pair.

- This eliminates the need for a second ARP request.

If machine A wishes to communicate with machine B, there is high probability that B will need A's Ethernet address as well.

Since every machine receives every ARP request (which is broadcast), how about adding the source address in each ARP request to the cache?

- Although a network may consist of hundreds of machines, a given host is unlikely to actively communicate with more than a few at any one time.
- Thus, adding every mapping to the local cache is likely to waste memory, and may cause the flushing of entries that will be used again soon to make room for entries that will never be used.





# ARP

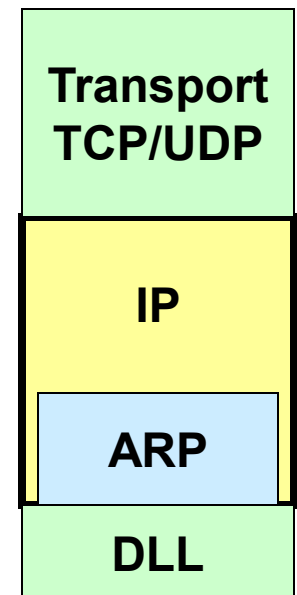
Solution:

Upon receipt of an ARP request from a machine whose IP address is already in the local ARP cache, update the information for that entry.

- This handles the case of a machine whose Ethernet address changes; ARP entries with the old value will be overwritten with the new value.

For a target on a remote network, it's a bit more complicated. Broadcasts don't cross routers. So, the requester, seeing that a request is remote, essentially needs to hand it off to a router to handle further.

From a layering point of view, ARP sits below IP, but above the data link layer.



## ARP Details

Conceptually, ARP consists of two parts: the software responsible for finding the physical address of an IP address (e.g., a client), and the software responsible for answering ARP requests from other machines (e.g., a server).

When sending an IP datagram, the sender searches its local ARP cache for the desired target address. If found, ARP is done.

If not found, send out a broadcast ARP request and wait for the response.

In practice, waiting for a response is somewhat tricky, because the target machine may be down, the request might become lost and need to be retransmitted, and so forth.

# ARP

ARP packets have been designed in a general way so that the protocol can be used over many different network technologies. ARP packets have the following format:

1. The 2-byte **Hardware-Type** field gives the type of the hardware address we are interested in (e.g., 1 for Ethernet).
2. The 2-byte **Protocol-Type** field gives the type of the higher level protocol address we are interested in (e.g., 0x0800 for IP). Note, it is two bytes long, just like the Ethernet type field.
3. A 1-byte **Hardware-Length** field specifying the length of the hardware address (6 bytes would be the length for Ethernet).
4. A 1-byte **Protocol-Length** field specifying the length of the target protocol address (4 for IP).
5. A 16-bit Operation Code field specifying the operation desired (e.g., REQUEST or RESPONSE).
6. The sender's Ethernet address (Sender Hardware Address) (if known).
7. The sender's Internet address (Sender Protocol Address) (if known).
8. The target's Ethernet address (Target Hardware Address) (filled in response).
9. The target's Internet address (Target Protocol Address) (filled in response).

# Reverse ARP

ARP handles the problem of determining the hardware address that corresponds to a given IP address.

But how do I find my own IP address? The protocol that maps hardware addresses to Internet addresses is called Reverse ARP, or RARP.

Necessary when a diskless machine first boots. It doesn't know its own IP address (and can't read it from a local disk!). The booting client contacts a server to obtain its Internet address.

1. The client communicates with a server by using a special protocol that requires only Ethernet frames. In essence it says "My ethernet address is aa.bb.cc.dd.ee.ff. Does anyone know my IP address?"
2. The broadcast goes to all nodes, including the RARP server. The RARP server maintains a database of physical address to Internet address mappings.
  - The actual format of RARP messages is similar to those of ARP:
  - The Ethernet frame type is set to type RARP (0x8035), and RARP defines two new message types; 'RARP request' and 'RARP response'.
  - The remaining fields are the same as in ARP.

We now see one of the primary benefits of broadcasting; locating servers.

However, because broadcasting is resource intensive (every machine on the local network must process the message, even if only to reject it).

# Dynamic Host Configuration Protocol (DHCP)

DHCP: Dynamic Host Configuration Protocol (RFC 1531)

Used to match workstations with an IP address. This address can be changed every time the machine boots. Allows configuration flexibility.

Here's the protocol:

1. Workstation broadcasts DHCPDISCOVER message on power-up.
2. Several DHCP Servers may respond with DHCPOFFER messages containing:  
IP address, subnet mask  
Router address  
Renewal Time
3. Workstation responds to one offer with DHCPREQUEST.  
Request may include items like: DNS servers, time servers, boot files,  
DHCP Server now binds IP address and replies with DHCPACK message with requested options.

Manager assigns multiple ranges of IP addresses to each DHCP server and server manages distribution to clients.

Client must renew IP address at regular intervals indicated by *Renewal Time*.

# Gateway Protocol

## **AS - Autonomous System:**

Those networks run by independent organizations (for instance, companies.)

Administrative regions that contain a set of networks and gateways.

A site is free to manage routing within its region any way it wishes, and routing information flows among regions only through carefully controlled mechanisms.

## **IGP - Interior Gateway Protocol:**

A routing protocol that's run within an AS.

1. ASs must be able to isolate themselves from other sites. They should be able to keep their local internets operating even when other parts of the Internet have failed.
2. Local gateways (probably) don't want to know (in much detail) about topological changes that take place far away.
3. Sites want administrative control over their gateways and networks and may not want to run the same routing protocols as other sites.

## **EGP - Exterior Gateway Protocol:**

A routing protocol that's run between ASs.

1. Allows a site to advertise to the rest of the world a path to the networks within its autonomous system.
2. Allows sites to learn about networks located in other autonomous regions.

# Interior Gateway Protocol - OSPF

## OSPF – Open Shortest Path First

Becoming the primary IGP. Allows an addressing hierarchy and thus makes routing easier.

The requirements used when designing OSPF included:

1. Had to be "Open" - published in the literature.
2. Had to support a number of "distance" metrics, including physical length, delay, capacity, etc.
3. Had to be dynamic, able to adapt to changing topologies.
4. Had to support "type of service" - able to change routing behavior based on frame characteristics.
5. Had to do load balancing; able to use multiple routes rather than one at a time.
6. Had to support hierarchical systems so that no one router needed to understand the entire flat network.
7. Had to provide some kind of security.

# Interior Gateway Protocol - OSPF

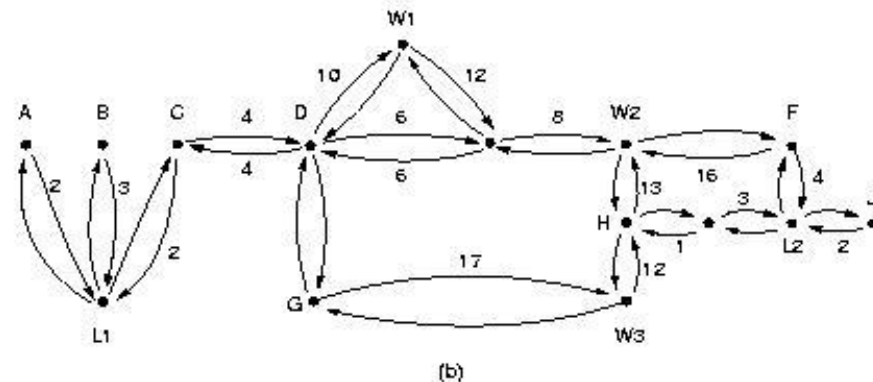
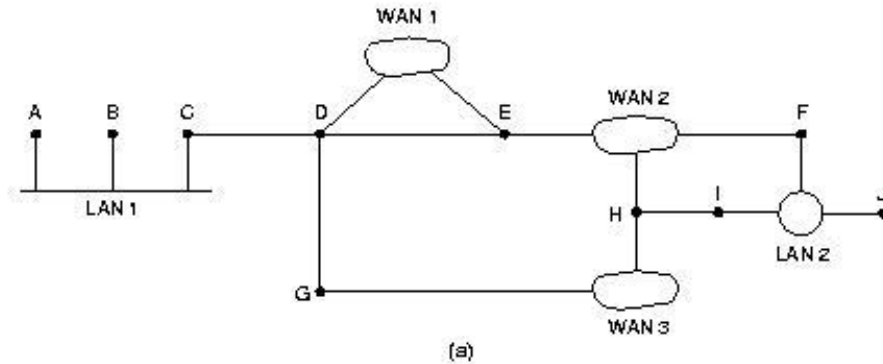
**OSPF** supports three kinds of networks:

1. Point to point lines between two routers.
2. Multiaccess networks with broadcasting (LANs).
3. Multiaccess networks without broadcasting (packet switched WANs ).

[Here a multiaccess network is one that has multiple routers, each of which can talk to all the other routers. This is a common LAN/WAN property.]

**As OSPF is defined, it:**

1. Divides an Autonomous System into “**areas**”. An area is a network or set of contiguous networks. All routers in an AS do not need to be in an Area.
2. Uses a link-state algorithm within an area. Thus distances are calculated based on length, or other properties.





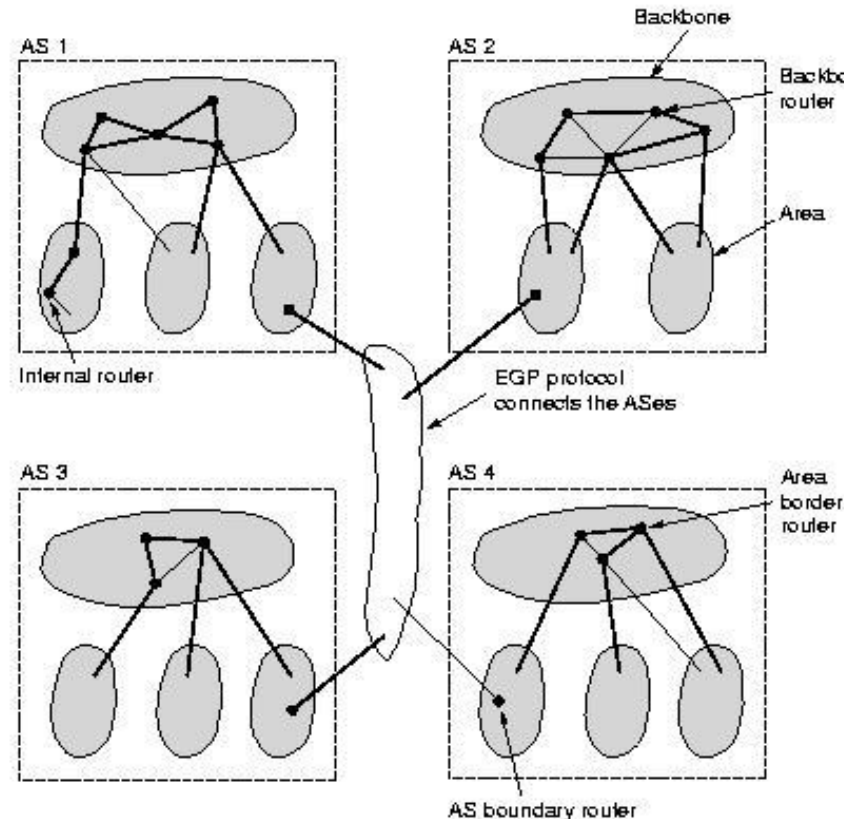
# Interior Gateway Protocol - OSPF

As OSPF is defined (continued), it:

3. Utilizes a Backbone. All areas are connected to the backbone so packets can travel from area to area via the backbone.

4. Employs four classes of routers see Figure

- Internal routers connecting networks wholly within one area.
- Backbone routers on the backbone area.
- Area border routers connecting two or more areas (includes connecting the backbone with an area.)
- AS boundary routers which talk to routers in other ASes.



# Interior Gateway Protocol - OSPF

## **As OSPF is defined (continued), it:**

5. Supports type of service routing. It provides for multiple paths, with gateways choosing paths based on the type of service field in IP headers.
6. Supports multipath routing. It distributes traffic over multiple paths to a destination.
7. Includes integrated support for subnetting. Specifically, (network number, network mask) pairs are distributed in updates.
8. Authenticates updates: Unauthenticated updates make the network extremely vulnerable to denial of service attacks (e.g., any workstation can send out bogus updates that break routing).

# Exterior Gateway Protocol - BGP

## **BORDER GATEWAY PROTOCOL (BGP)**

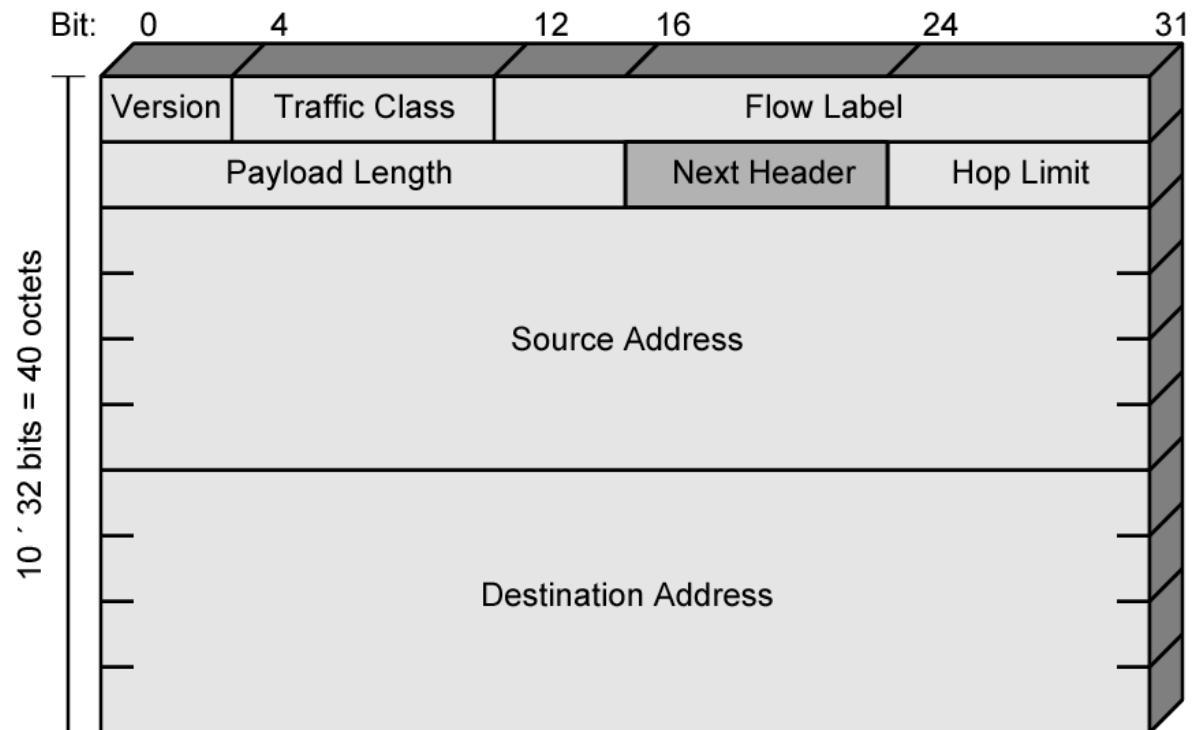
BGP is the current Exterior Gateway Routing Protocol ( EGP ) used.

Distance vector protocol, but not only does it account for distance, but also for specific route criteria.

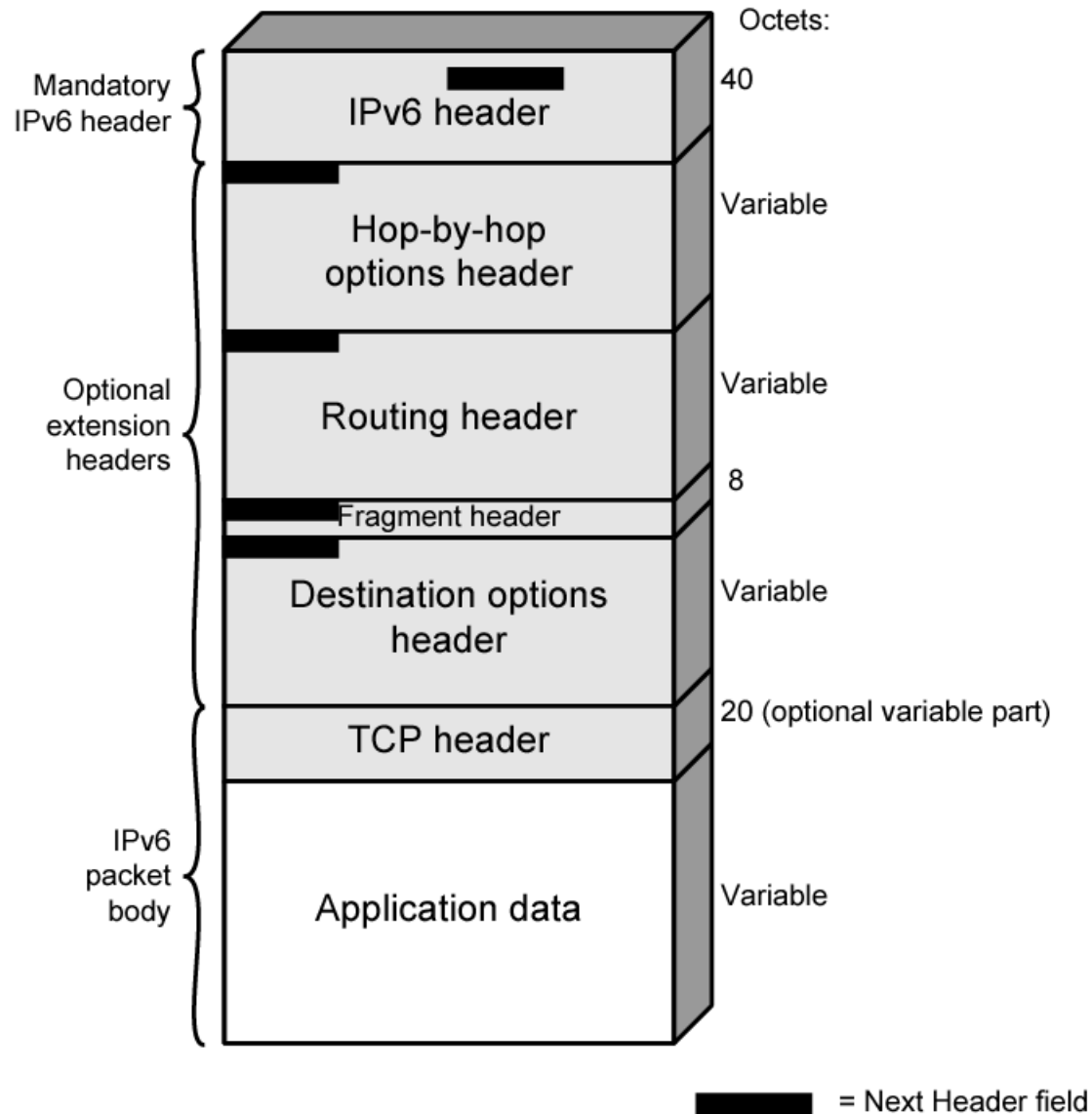
BGP can take into account politics, security and economic issues.

# IPv6

- **Version**
  - 6
- **Traffic Class**
  - Classes or priorities of packet
  - Still under development
  - See RFC 2460
- **Flow Label**
  - Used by hosts requesting special handling
- **Payload length**
  - Includes all extension headers plus user data
- **Next Header**
  - Identifies type of header
    - Extension or next layer up
- **Source Address (16 bytes)**
- **Destination address (16 bytes)**



# IPv6 Structure



## Some Useful Tools

**Netstat** - tells you about the connections you have open on your machine.

**Ping** - tells you how long it takes to get to a destination (and if there is a route to that destination).

**Arp** - gives information about the routing table.

**Finger** - tells you who is logged on.

**ftp** - gets you data from a remote site.

**Route** - tells you information about the routing tables.

**Telnet** – allows you to log on to a remote host.

**Tracert** – Find the paths to remote sites. A useful site is [www.traceroute.org](http://www.traceroute.org)